

GATE Data Science & Artificial Intelligence (DA)

Comprehensive 80-Question Topic-Wise Question Bank

10 PYQs + 10 Practice Questions with Detailed Step-by-Step Solutions per Topic

Aligned with the official GATE DA Syllabus: Machine Learning, DBMS, Probability & Statistics, and Aptitude

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1 Topic 1: Statistical Regression & Correlation Coefficient

1.1 10 Previous Year Questions (PYQs)

Q1.1

[GATE DA 2024 | NAT | 2 Marks]

For a sample of $n = 5$ data points, the sample means are $\bar{x} = 4$ and $\bar{y} = 8$. The sample variances are $s_x^2 = 4$ and $s_y^2 = 9$, and the sample covariance is $\text{Cov}(x, y) = 4.8$. What is the value of the sample Pearson correlation coefficient r_{xy} ? **Correct Answer: 0.80** **Solution:** $r_{xy} = \frac{\text{Cov}(x, y)}{s_x s_y} = \frac{4.8}{\sqrt{4}\sqrt{9}} = \frac{4.8}{2 \times 3} = \frac{4.8}{6} = 0.80$.

Q1.2

[GATE DA 2024 | MCQ | 1 Mark]

Let $Y = \beta_0 + \beta_1 X + \epsilon$ be a simple linear regression model. If r denotes the Pearson correlation coefficient between X and Y , which of the following statements is ALWAYS true?

- (A) β_1 and r must have identical signs.
- (B) r^2 is strictly greater than the coefficient of determination R^2 .
- (C) If $r = 0$, then X and Y are necessarily statistically independent.
- (D) $\beta_1 = r$.

Correct Answer: (A) **Solution:** Since $\hat{\beta}_1 = r \frac{s_y}{s_x}$ and the sample standard deviations $s_x, s_y > 0$, $\hat{\beta}_1$ has the exact same algebraic sign as r . (C) is false because correlation only captures linear dependence.

Q1.3

[GATE ST 2023 | NAT | 2 Marks]

Let the two regression lines for a bivariate dataset be:

$$2y - x - 50 = 0 \quad \text{and} \quad 3x - 2y - 10 = 0$$

What is the value of the correlation coefficient r_{xy} ? (Round off to two decimal places) **Correct Answer: 0.58** **Solution:** Write regressions in standard forms: y on $x \implies y = 0.5x + 25 \implies b_{yx} = 0.5$. x on $y \implies x = \frac{2}{3}y + \frac{10}{3} \implies b_{xy} = \frac{2}{3}$. Since $r^2 = b_{yx}b_{xy} = 0.5 \times \frac{2}{3} = \frac{1}{3} \approx 0.3333$, we have $r = +\sqrt{1/3} \approx 0.58$.

Q1.4

[GATE DA 2025 | MSQ | 2 Marks]

Let $U = aX + b$ and $V = cY + d$, where a, b, c, d are real constants with $a \neq 0$ and $c \neq 0$. Let $r(X, Y)$ denote the Pearson correlation coefficient. Which of the following statements is/are TRUE?

- (A) If $a > 0$ and $c > 0$, then $r(U, V) = r(X, Y)$.
- (B) If $a > 0$ and $c < 0$, then $r(U, V) = -r(X, Y)$.
- (C) If $a < 0$ and $c < 0$, then $r(U, V) = r(X, Y)$.
- (D) $r(U, V)$ depends on the shift constants b and d .

Correct Answer: (A, B, C) **Solution:** By invariance of correlation under affine transformations, $r(aX + b, cY + d) = \frac{ac}{|ac|} r(X, Y) = \text{sign}(ac) r(X, Y)$. Shift constants b and d cancel out completely.

Q1.5

[GATE ST 2022 | MCQ | 1 Mark]

If the angle between the two sample regression lines of Y on X and X on Y is 0° , then the correlation coefficient r must be:

- (A) 0
- (B) +1 or -1

(C) 0.5

(D) Undefined

Correct Answer: (B) **Solution:** The angle θ between the two regression lines satisfies $\tan \theta = \frac{1-r^2}{|r|} \left(\frac{s_x s_y}{s_x^2 + s_y^2} \right)$. If $\theta = 0$, then $\tan \theta = 0 \implies 1 - r^2 = 0 \implies r = \pm 1$. The two lines coincide perfectly.

Q1.6

[GATE DA 2024 | NAT | 2 Marks]

In a simple linear regression analysis with $n = 20$ observations, the total sum of squares is $SST = 250$ and the error sum of squares is $SSE = 40$. The value of the sample correlation coefficient r_{xy} satisfies $r_{xy} > 0$. What is the value of r_{xy} ? (Round off to two decimal places) **Correct Answer:**

0.92 **Solution:** $R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{40}{250} = 1 - 0.16 = 0.84$. In simple bivariate regression, $R^2 = r_{xy}^2 \implies r_{xy} = \sqrt{0.84} \approx 0.9165 \approx 0.92$.

Q1.7

[GATE ST 2021 | MCQ | 2 Marks]

Let X and Y be two zero-mean random variables with $\text{Var}(X) = 1$, $\text{Var}(Y) = 4$, and $\text{Cov}(X, Y) = 1$. The slope of the best linear predictor of Y given X without intercept is:

(A) 0.25

(B) 0.50

(C) 1.00

(D) 2.00

Correct Answer: (C) **Solution:** The optimal least squares slope is $\beta = \frac{\text{Cov}(X, Y)}{\text{Var}(X)} = \frac{1}{1} = 1.00$.

Q1.8

[GATE DA 2025 | NAT | 1 Mark]

If the regression coefficient of Y on X is $b_{yx} = -0.64$ and the regression coefficient of X on Y is $b_{xy} = -0.25$, then the correlation coefficient r_{xy} is _____. **Correct Answer: -0.40** **Solution:** $r_{xy} = -\sqrt{b_{yx} b_{xy}} = -\sqrt{(-0.64)(-0.25)} = -\sqrt{0.16} = -0.40$. The sign of r must match both regression coefficients.

Q1.9

[GATE ST 2020 | MSQ | 2 Marks]

For random variables X and Y , which of the following statements is/are always TRUE?

(A) $-1 \leq \text{Corr}(X, Y) \leq 1$.

(B) If $\text{Corr}(X, Y) = 0$, then $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$.

(C) If $Y = X^2$ where $X \sim \mathcal{N}(0, 1)$, then $\text{Corr}(X, Y) = 0$.

(D) If $\text{Corr}(X, Y) = 1$, then $Y = aX + b$ with $a > 0$.

Correct Answer: (A, B, C, D) **Solution:** All four are true. (A) Cauchy-Schwarz inequality. (B) $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$, and $\text{Cov} = 0$. (C) $\text{Cov}(X, X^2) = E[X^3] - E[X]E[X^2] = 0 - 0 = 0$. (D) $r = 1$ indicates an exact positive linear deterministic relationship.

Q1.10

[GATE DA 2026 | NAT | 2 Marks]

Given $\sum_{i=1}^{10} (x_i - 5)^2 = 40$, $\sum_{i=1}^{10} (y_i - 10)^2 = 90$, and $\sum_{i=1}^{10} (x_i - 5)(y_i - 10) = 48$, where $\bar{x} = 5$ and $\bar{y} = 10$. What is the value of the sample correlation coefficient r ? **Correct Answer: 0.80**

Solution: $r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} = \frac{48}{\sqrt{40 \times 90}} = \frac{48}{\sqrt{3600}} = \frac{48}{60} = 0.80$.

1.2 10 Practice Questions (GATE Level)

P1.1

[Practice | MCQ | 1 Mark]

If $r_{xy} = 0.6$, what percentage of the total variation in the dependent variable Y is explained by the independent variable X in a simple linear regression?

- (A) 60%
- (B) 36%
- (C) 40%
- (D) 64%

Correct Answer: (B) **Solution:** The proportion of explained variance is given by the coefficient of determination $R^2 = r_{xy}^2 = (0.6)^2 = 0.36 = 36\%$.

P1.2

[Practice | NAT | 2 Marks]

Let $\hat{y} = 3 + 2x$ be the regression line of y on x , and $\hat{x} = -1 + 0.32y$ be the regression line of x on y . Compute the Pearson correlation coefficient between x and y . **Correct Answer: 0.80** **Solution:** $b_{yx} = 2$ and $b_{xy} = 0.32$. Both are positive, so $r = +\sqrt{b_{yx}b_{xy}} = \sqrt{2 \times 0.32} = \sqrt{0.64} = 0.80$.

P1.3

[Practice | MSQ | 2 Marks]

Which of the following conditions guarantee that the ordinary least squares slope $\hat{\beta}_1$ in simple linear regression equals 1?

- (A) $\text{Cov}(x, y) = \text{Var}(x)$.
- (B) $r_{xy} = 1$ and $s_x = s_y$.
- (C) $r_{xy} = 0.5$ and $s_y = 2s_x$.
- (D) $\text{Var}(y) = \text{Var}(x)$.

Correct Answer: (A, B, C) **Solution:** $\hat{\beta}_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)} = r_{xy} \frac{s_y}{s_x}$. (A) directly yields 1. (B) $1 \times 1 = 1$. (C) $0.5 \times 2 = 1$. (D) is insufficient without knowledge of r_{xy} .

P1.4

[Practice | NAT | 1 Mark]

A dataset has 6 pairs (x_i, y_i) . If $\sum (x_i - \bar{x})^2 = 50$, $\sum (y_i - \bar{y})^2 = 50$, and the residual sum of squares is $\text{SSE} = 18$, calculate $|r_{xy}|$. **Correct Answer: 0.80** **Solution:** $R^2 = 1 - \frac{\text{SSE}}{\text{SST}} = 1 - \frac{18}{50} = 1 - 0.36 = 0.64$. Since $R^2 = r^2$, $|r_{xy}| = \sqrt{0.64} = 0.80$.

P1.5

[Practice | MCQ | 1 Mark]

Suppose the correlation between variables X and Y is $r = -0.7$. If every X value is multiplied by -3 and 5 is subtracted from every Y value, what is the new correlation?

- (A) -0.7
- (B) 0.7
- (C) -0.35
- (D) 0.35

Correct Answer: (B) **Solution:** Let $U = -3X$ and $V = Y - 5$. Here $a = -3 < 0$ and $c = 1 > 0$. The sign of ac is negative, so $r(U, V) = \text{sign}(ac)r(X, Y) = (-1)(-0.7) = +0.7$.

P1.6

[Practice | NAT | 2 Marks]

For a sample of observations, the mean of X is 10 and the mean of Y is 20. The regression line of Y on X passes through $(10, 20)$ and $(15, 35)$. What is the estimated value of $\hat{Y}$ when $X = 12$?

Correct Answer: 26 **Solution:** The slope is $\hat{\beta}_1 = \frac{35-20}{15-10} = \frac{15}{5} = 3$. The equation is $\hat{Y} - 20 = 3(X - 10) \implies \hat{Y} = 3X - 10$. For $X = 12$, $\hat{Y} = 3(12) - 10 = 26$.

P1.7

[Practice | MCQ | 2 Marks]

Under the classical linear regression model $y_i = \beta_0 + \beta_1 x_i + \epsilon_i$ with Gauss-Markov assumptions ($\epsilon_i \sim \text{i.i.d.}(0, \sigma^2)$), the variance of the OLS estimator $\hat{\beta}_1$ is:

- (A) $\frac{\sigma^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$
- (B) $\frac{\sigma^2}{n}$
- (C) $\frac{\sigma^2 \sum x_i^2}{n \sum (x_i - \bar{x})^2}$
- (D) $\frac{\sigma}{\sqrt{n}}$

Correct Answer: (A) **Solution:** By standard derivation, $\text{Var}(\hat{\beta}_1) = \text{Var}(\sum w_i y_i) = \sum w_i^2 \sigma^2 = \frac{\sigma^2}{\sum (x_i - \bar{x})^2}$.

P1.8

[Practice | MSQ | 2 Marks]

Which of the following statements about simple linear regression residuals $e_i = y_i - \hat{y}_i$ is/are TRUE when an intercept is included?

- (A) $\sum_{i=1}^n e_i = 0$
- (B) The sample correlation between e_i and the fitted values $\hat{y}_i$ is strictly zero.
- (C) The sample correlation between e_i and feature values x_i is strictly zero.
- (D) $\sum_{i=1}^n e_i^2 = \sum_{i=1}^n y_i^2 - \sum_{i=1}^n \hat{y}_i^2$

Correct Answer: (A, B, C, D) **Solution:** All are exact algebraic properties: (A) follows from $\partial J / \partial \beta_0 = 0$. (B) and (C) follow from orthogonality conditions $\mathbf{e}^T \hat{\mathbf{y}} = 0$ and $\mathbf{e}^T \mathbf{x} = 0$. (D) follows from Pythagoras' theorem on the orthogonal decomposition $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{e}$.

P1.9

[Practice | NAT | 1 Mark]

If $s_x = 4$, $s_y = 6$, and the regression coefficient of y on x is $b_{yx} = 1.2$, what is the regression coefficient of x on y (b_{xy})? **Correct Answer: 0.533** **Solution:** $b_{yx} = r \frac{s_y}{s_x} \implies 1.2 = r \frac{6}{4} = 1.5r \implies r = \frac{1.2}{1.5} = 0.8$. Then $b_{xy} = r \frac{s_x}{s_y} = 0.8 \times \frac{4}{6} = \frac{3.2}{6} \approx 0.533$.

P1.10

[Practice | MCQ | 1 Mark]

Spearman's rank correlation coefficient between two variables is computed on:

- (A) The original raw quantitative observations.
- (B) The monotonic ranks assigned to the values of each variable.
- (C) The residuals of a polynomial fit.
- (D) The normalized z-scores without ordering.

Correct Answer: (B) **Solution:** Spearman's ρ is precisely Pearson's correlation applied to the ordinal ranks of the two variables, assessing monotonic relationships.

2 Topic 2: Boosting: AdaBoost & Gradient Boosting

2.1 10 Previous Year Questions (PYQs)

Q2.1

[GATE DA 2024 | MCQ | 1 Mark]

In AdaBoost (Adaptive Boosting) for binary classification with targets $y_i \in \{-1, +1\}$, if a weak base learner achieves a weighted error rate of $\epsilon_m = 0.5$, what is its assigned learner weight α_m ?

- (A) 1.0
- (B) 0.0
- (C) 0.5
- (D) -1.0

Correct Answer: (B) **Solution:** In AdaBoost, $\alpha_m = \frac{1}{2} \ln \left(\frac{1-\epsilon_m}{\epsilon_m} \right)$. When $\epsilon_m = 0.5$, $\alpha_m = \frac{1}{2} \ln(1) = 0$. A base classifier performing at chance level is assigned zero voting weight.

Q2.2

[GATE DA 2024 | NAT | 2 Marks]

In AdaBoost with binary labels $y \in \{-1, +1\}$, a weak learner produces an error rate $\epsilon = 0.2$. What is the factor by which the weight of an incorrectly classified sample is multiplied relative to a correctly classified sample? **Correct Answer: 4.00** **Solution:** Correct samples are scaled by $e^{-\alpha}$ and misclassified samples by $e^{+\alpha}$. The relative multiplier is $\frac{e^{\alpha}}{e^{-\alpha}} = e^{2\alpha} = \exp \left(2 \cdot \frac{1}{2} \ln \left(\frac{1-\epsilon}{\epsilon} \right) \right) = \frac{1-\epsilon}{\epsilon} = \frac{1-0.2}{0.2} = \frac{0.8}{0.2} = 4.00$.

Q2.3

[GATE DA 2025 | MSQ | 2 Marks]

Which of the following statements comparing Bagging and Boosting is/are TRUE?

- (A) Bagging builds base learners independently in parallel, whereas Boosting builds them sequentially.
- (B) Bagging primarily reduces variance, whereas Boosting primarily reduces bias.
- (C) AdaBoost is generally more sensitive to noisy data and outliers than Random Forest.
- (D) Boosting trains each base estimator on a completely unweighted uniform bootstrap sample.

Correct Answer: (A, B, C) **Solution:** (A) and (B) are core ensemble learning properties. (C) is true because AdaBoost continually inflates weights of stubborn misclassified points (including outliers). (D) is false; AdaBoost adaptively reweights training examples.

Q2.4

[GATE DA 2025 | MCQ | 1 Mark]

In Gradient Boosting for regression using the squared error loss $L(y, F(x)) = \frac{1}{2}(y - F(x))^2$, what target does the m -th tree fit to at iteration m ?

- (A) The original targets y_i .
- (B) The raw residuals $y_i - F_{m-1}(x_i)$.
- (C) The ratio $y_i / F_{m-1}(x_i)$.
- (D) The exponential loss $e^{-y_i F_{m-1}(x_i)}$.

Correct Answer: (B) **Solution:** Gradient boosting fits base learners to the pseudo-residuals: $r_{im} = - \left[\frac{\partial L(y_i, F(x_i))}{\partial F(x_i)} \right]_{F=F_{m-1}} = y_i - F_{m-1}(x_i)$, which are exactly the raw residuals under squared error loss.

Q2.5

[GATE CS 2022 | MCQ | 2 Marks]

AdaBoost can be mathematically viewed as minimizing which of the following objective loss functions using forward stagewise additive modeling?

- (A) Exponential loss: $L(y, f(x)) = \exp(-yf(x))$
- (B) Hinge loss: $L(y, f(x)) = \max(0, 1 - yf(x))$
- (C) Cross-entropy loss: $L(y, f(x)) = \log(1 + e^{-yf(x)})$
- (D) Mean absolute error: $L(y, f(x)) = |y - f(x)|$

Correct Answer: (A) **Solution:** As demonstrated by Friedman, Hastie, and Tibshirani (2000), AdaBoost sequentially minimizes the empirical exponential loss $\sum_{i=1}^n \exp(-y_i f(x_i))$.

Q2.6

[GATE DA 2026 | NAT | 2 Marks]

In AdaBoost, there are 10 training samples initially assigned equal weights $w_i^{(1)} = 0.1$. The first weak classifier misclassifies 2 samples. What is the weighted training error ϵ_1 ? **Correct Answer:**

0.20 **Solution:** $\epsilon_1 = \sum_{i: y_i \neq h_1(x_i)} w_i^{(1)} = 0.1 + 0.1 = 0.20$.

Q2.7

[GATE DA 2025 | NAT | 2 Marks]

In Gradient Boosting with learning rate (shrinkage) $\nu = 0.1$, the initial model predicts $F_0(x) = 10.0$ for all x . For an input point with true label $y = 15.0$, the first decision tree fits the residual and predicts $h_1(x) = 4.5$. What is the updated prediction $F_1(x)$? **Correct Answer: 10.45** **Solution:** The additive update rule is $F_1(x) = F_0(x) + \nu \cdot h_1(x) = 10.0 + 0.1 \times 4.5 = 10.0 + 0.45 = 10.45$.

Q2.8

[GATE CS 2023 | MCQ | 1 Mark]

What is the typical base estimator used in standard AdaBoost and Gradient Tree Boosting?

- (A) Support Vector Machines with RBF kernel
- (B) Shallow Decision Trees (such as decision stumps)
- (C) Multi-layer perceptrons with 5 hidden layers
- (D) Fully grown deep unpruned trees

Correct Answer: (B) **Solution:** Boosting sequentially combines high-bias, low-variance weak learners. Shallow decision trees (depth 1 to 3) or decision stumps are the canonical base estimators.

Q2.9

[GATE DA 2024 | MSQ | 2 Marks]

Which of the following techniques can prevent overfitting in Gradient Boosted Decision Trees (GBDT)?

- (A) Lowering the learning rate ν (shrinkage) combined with early stopping.
- (B) Subsampling rows (Stochastic Gradient Boosting).
- (C) Restricting the maximum depth of individual trees.
- (D) Increasing the number of boosting iterations to infinity.

Correct Answer: (A, B, C) **Solution:** (A), (B), and (C) are standard regularization techniques in GBDT. (D) increases model complexity and leads directly to overfitting on training noise.

Q2.10

[GATE CS 2021 | NAT | 1 Mark]

If an AdaBoost classifier combines 3 base learners h_1, h_2, h_3 with weights $\alpha_1 = 0.8$, $\alpha_2 = 0.5$, and $\alpha_3 = 0.4$. For an input x , the predictions are $h_1(x) = -1$, $h_2(x) = +1$, and $h_3(x) = +1$. The final ensemble prediction $\text{sign}\left(\sum_{m=1}^3 \alpha_m h_m(x)\right)$ is $+1$ or -1 ? (Enter $+1$ or -1) **Correct Answer: +1** **Solution:** Compute the weighted voting sum: $S = \alpha_1 h_1(x) + \alpha_2 h_2(x) + \alpha_3 h_3(x) = 0.8(-1) + 0.5(+1) + 0.4(+1) = -0.8 + 0.9 = +0.1$. Since $S > 0$, the ensemble predicts $\text{sign}(+0.1) = +1$.

2.2 10 Practice Questions (GATE Level)

P2.1

[Practice | MCQ | 1 Mark]

In AdaBoost, what happens to the weight α_m of a classifier whose weighted error rate is $\epsilon_m > 0.5$?

- (A) $\alpha_m > 0$
- (B) $\alpha_m < 0$
- (C) α_m is undefined
- (D) $\alpha_m = 0.5$

Correct Answer: (B) **Solution:** $\alpha_m = \frac{1}{2} \ln \left(\frac{1-\epsilon_m}{\epsilon_m} \right)$. If $\epsilon_m > 0.5$, then $\frac{1-\epsilon_m}{\epsilon_m} < 1$, so the natural logarithm is negative, yielding $\alpha_m < 0$ (which effectively inverts the classifier's decisions).

P2.2

[Practice | NAT | 2 Marks]

For a weak classifier with error rate $\epsilon_m = \frac{1}{1+e}$, calculate the classifier weight α_m . (Use $e \approx 2.718$, $\ln e = 1$) **Correct Answer: 0.50** **Solution:** $\frac{1-\epsilon_m}{\epsilon_m} = \frac{1-\frac{1}{1+e}}{\frac{1}{1+e}} = \frac{e}{1+e} = e$. Therefore, $\alpha_m = \frac{1}{2} \ln(e) = \frac{1}{2}(1) = 0.50$.

P2.3

[Practice | MSQ | 2 Marks]

In Gradient Boosting for regression with absolute error loss $L(y, F(x)) = |y - F(x)|$, the pseudo-residuals $r_i = -\frac{\partial L}{\partial F}$ are given by:

- (A) $r_i = \text{sign}(y_i - F(x_i))$ for $y_i \neq F(x_i)$.
- (B) The base tree fits the sign of the error rather than the magnitude.
- (C) It produces a model equivalent to median regression (quantile regression with $\alpha = 0.5$).
- (D) It is extremely sensitive to large outliers.

Correct Answer: (A, B, C) **Solution:** The derivative of $|y - F|$ with respect to F is $-\text{sign}(y - F)$, so the negative gradient is $+\text{sign}(y - F)$. This leads to robust median regression, which is resilient to outliers (making D false).

P2.4

[Practice | NAT | 2 Marks]

In AdaBoost, after updating the sample weights $w_i \leftarrow w_i \exp(-\alpha y_i h(x_i))$, the normalizer is $Z = \sum_{i=1}^n w_i$. If the error rate was $\epsilon = 0.1$, what is the theoretical value of $Z = 2\sqrt{\epsilon(1-\epsilon)}$? (Round off to two decimal places) **Correct Answer: 0.60** **Solution:** $Z = 2\sqrt{\epsilon(1-\epsilon)} = 2\sqrt{0.1 \times 0.9} = 2\sqrt{0.09} = 2 \times 0.3 = 0.60$.

P2.5

[Practice | MCQ | 1 Mark]

Which of the following describes XGBoost's primary architectural algorithmic advancement over standard Gradient Boosting?

- (A) Uses second-order Taylor expansion (Newton-Raphson approximation) of the loss function with explicit tree leaf regularization.
- (B) Replaces decision trees with multi-layer neural networks.
- (C) Trains all trees independently using bootstrap aggregation.
- (D) Ignores gradients and uses genetic algorithms.

Correct Answer: (A) **Solution:** XGBoost incorporates both first and second-order gradients (g_i and h_i) via second-order Taylor expansion and adds L_1 and L_2 penalties directly on leaf weights.

P2.6

[Practice | NAT | 2 Marks]

Consider an ensemble of 5 independent binary classifiers, each with individual accuracy $p = 0.70$. What is the majority voting ensemble accuracy? (Round off to three decimal places) **Correct**

Answer: 0.837 **Solution:** Majority vote requires at least 3 out of 5 classifiers to be correct:
 $P(K \geq 3) = \binom{5}{3}(0.7)^3(0.3)^2 + \binom{5}{4}(0.7)^4(0.3)^1 + \binom{5}{5}(0.7)^5 = 10(0.343)(0.09) + 5(0.2401)(0.3) + 0.16807 = 0.3087 + 0.36015 + 0.16807 = 0.83692 \approx 0.837$.

P2.7

[Practice | MCQ | 1 Mark]

Why does Gradient Boosting use a shrinkage parameter (learning rate $\nu \in (0, 1]$)?

- (A) To prevent individual trees from dominating the ensemble, reducing overfitting.
- (B) To make the algorithm run faster with fewer trees.
- (C) To ensure all training residuals become exactly zero after the first step.
- (D) To eliminate the need for computing loss gradients.

Correct Answer: (A) **Solution:** Shrinkage scales each step by $\nu < 1$, forcing the model to learn gradually from the residuals of many trees, which significantly improves generalization.

P2.8

[Practice | MSQ | 2 Marks]

Which of the following statements regarding AdaBoost sample weight updates is/are TRUE?

- (A) The total sum of weights of correctly classified samples after reweighting (before normalization) equals $\frac{1}{2}$.
- (B) The total sum of weights of misclassified samples after reweighting equals the sum of weights of correctly classified samples.
- (C) The weighted error rate on the updated distribution using the same weak classifier h_m is exactly 0.5.
- (D) Correctly classified points receive decreased normalized weights.

Correct Answer: (B, C, D) **Solution:** By design of AdaBoost, updating and renormalizing ensures that the weighted error of h_m under the new weights $w^{(m+1)}$ is exactly 0.5, meaning correct and incorrect samples each carry half the total weight.

P2.9

[Practice | NAT | 2 Marks]

In AdaBoost, if $\epsilon_m = 0.1192$, then $\frac{1-\epsilon_m}{\epsilon_m} \approx 7.389 = e^2$. What is the value of α_m ? **Correct Answer:**

1.00 **Solution:** $\alpha_m = \frac{1}{2} \ln \left(\frac{1-\epsilon_m}{\epsilon_m} \right) = \frac{1}{2} \ln(e^2) = \frac{1}{2} \times 2 = 1.00$.

P2.10

[Practice | MCQ | 1 Mark]

In Gradient Boosting for multi-class classification with K classes, how many decision trees are constructed in each boosting iteration?

- (A) 1
- (B) K
- (C) $K - 1$
- (D) 2^K

Correct Answer: (B) **Solution:** Standard gradient boosting for multi-class classification trains K separate trees per iteration (one for each class's softmax probability gradient).

3 Topic 3: Database Normalization: 1NF, 2NF, 3NF

3.1 10 Previous Year Questions (PYQs)

Q3.1

[GATE CS 2024 | MCQ | 1 Mark]

A relation R is in First Normal Form (1NF) if and only if:

- (A) Every non-prime attribute is fully functionally dependent on every candidate key.
- (B) All attribute domains contain only atomic (indivisible) values.
- (C) There are no transitive functional dependencies.
- (D) Every determinant is a candidate key.

Correct Answer: (B) **Solution:** 1NF strictly requires that the domain of each attribute contains only indivisible, atomic values, and that there are no repeating groups or composite attributes.

Q3.2

[GATE CS 2023 | NAT | 2 Marks]

Consider relation $R(A, B, C, D)$ with functional dependencies:

$$F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$$

What is the number of candidate keys for R ? **Correct Answer: 1** **Solution:** Compute closures: $A^+ = \{A, B, C, D\}$. $B^+ = \{B, C, D\}$. $C^+ = \{C, D\}$. $D^+ = \{D\}$. Only A determines all attributes, so A is the unique candidate key.

Q3.3

[GATE DA 2024 | MCQ | 2 Marks]

Consider relation $R(A, B, C, D)$ with candidate key AB and functional dependencies:

$$F = \{AB \rightarrow C, B \rightarrow D\}$$

What is the highest normal form satisfied by relation R ?

- (A) 1NF but not 2NF
- (B) 2NF but not 3NF
- (C) 3NF
- (D) BCNF

Correct Answer: (A) **Solution:** The candidate key is AB . Prime attributes: $\{A, B\}$. Non-prime attributes: $\{C, D\}$. In $B \rightarrow D$, D (non-prime) depends on B , which is a proper subset of candidate key AB . This partial dependency violates 2NF. Hence, R is in 1NF only.

Q3.4

[GATE CS 2022 | NAT | 1 Mark]

Consider relation $R(P, Q, R, S)$ with $F = \{P \rightarrow Q, Q \rightarrow R, R \rightarrow S, S \rightarrow P\}$. How many candidate keys does R have? **Correct Answer: 4** **Solution:** Closures: $P^+ = PQRS$, $Q^+ = QRSP$, $R^+ = RSPQ$, $S^+ = SPQR$. Each of the 4 single attributes forms a valid candidate key.

Q3.5

[GATE DA 2025 | MSQ | 2 Marks]

A relation R is in Third Normal Form (3NF) if for every non-trivial functional dependency $X \rightarrow Y$:

- (A) X is a superkey of R , OR
- (B) Y is a prime attribute (part of some candidate key) of R .
- (C) There are no partial dependencies on any candidate key.
- (D) Every multivalued dependency is a functional dependency.

Correct Answer: (A, B, C) **Solution:** By Codd and Zaniolo's definition of 3NF, every non-trivial dependency must have X as a superkey or Y as a prime attribute. (C) is also true because any 3NF relation is automatically in 2NF. (D) belongs to 4NF.

Q3.6

[GATE CS 2021 | MCQ | 2 Marks]

Relation $R(A, B, C, D, E)$ has functional dependencies $F = \{A \rightarrow B, BC \rightarrow D, E \rightarrow C\}$. The candidate key(s) of R is/are:

- (A) AE
- (B) AB
- (C) ACE
- (D) ADE

Correct Answer: (A) **Solution:** Check right-hand sides: A and E never appear on the right side of any dependency, so every candidate key MUST contain AE . Now check closure: $(AE)^+ \Rightarrow A \rightarrow B$ gives ABE ; $E \rightarrow C$ gives $ABCE$; $BC \rightarrow D$ gives $ABCDE$. Thus AE is the unique minimal candidate key.

Q3.7

[GATE CS 2020 | NAT | 2 Marks]

Let $R(A, B, C, D, E, F)$ have functional dependencies $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow E, E \rightarrow F, F \rightarrow A\}$. The candidate key of R is B combined with which single attributes? What is the total count of candidate keys? **Correct Answer: 5** **Solution:** Since B never appears on any RHS, B must belong to every candidate key. Starting from $AB \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow A$, the cycle $\{A, C, D, E, F\}$ allows each of these 5 attributes when paired with B to determine all attributes. Thus, the candidate keys are AB, CB, DB, EB, FB (total 5).

Q3.8

[GATE DA 2026 | MCQ | 1 Mark]

A relation $R(A, B, C)$ has functional dependencies $F = \{A \rightarrow B, B \rightarrow C\}$. Candidate key is A . The decomposition of R into $R_1(A, B)$ and $R_2(B, C)$ is:

- (A) Lossless and dependency preserving
- (B) Lossy and dependency preserving
- (C) Lossless but not dependency preserving
- (D) Neither lossless nor dependency preserving

Correct Answer: (A) **Solution:** Intersection is $R_1 \cap R_2 = \{B\}$. In R_2 , $B \rightarrow C$ holds, so B is a key of R_2 , ensuring a lossless join. $A \rightarrow B$ is preserved in R_1 and $B \rightarrow C$ in R_2 . Thus it is both lossless and dependency preserving.

Q3.9

[GATE CS 2019 | MCQ | 2 Marks]

Relation $R(A, B, C, D)$ has $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D, D \rightarrow A\}$. In what highest normal form is R ?

- (A) 1NF
- (B) 2NF
- (C) 3NF
- (D) BCNF

Correct Answer: (D) **Solution:** Closures of A, B, C, D each generate $\{A, B, C, D\}$. Thus A, B, C, D are all individual candidate keys. In every dependency $X \rightarrow Y$, the left-hand side is a candidate key (superkey). Therefore, R is in BCNF (and trivially 3NF).

Q3.10**[GATE DA 2024 | NAT | 1 Mark]**

For a relation $R(A, B, C, D)$ with candidate key AB , how many non-prime attributes exist if no other candidate keys exist? **Correct Answer: 2** **Solution:** Total attributes = 4. Prime attributes (part of candidate key AB) = $\{A, B\}$ (2 attributes). Non-prime attributes = $\{C, D\}$ (2 attributes).

3.2 10 Practice Questions (GATE Level)**P3.1****[Practice | MCQ | 1 Mark]**

A relation R with candidate key K is in 2NF if and only if:

- (A) It is in 1NF and no non-prime attribute is partially dependent on any candidate key.
- (B) Every determinant is a candidate key.
- (C) No prime attribute is transitively dependent on a candidate key.
- (D) All functional dependencies have atomic right-hand sides.

Correct Answer: (A) **Solution:** 2NF requires 1NF and the absence of partial dependencies (no non-prime attribute may depend on a proper subset of any candidate key).

P3.2**[Practice | NAT | 2 Marks]**

Consider $R(A, B, C, D, E)$ with $F = \{A \rightarrow BC, CD \rightarrow E, B \rightarrow D, E \rightarrow A\}$. How many candidate keys exist? **Correct Answer: 2** **Solution:** Check attributes: none are missing from both sides. Test A : $A^+ = ABC$, with $B \rightarrow D \implies ABCD$, with $CD \rightarrow E \implies ABCDE$. So A is a key. Since $E \rightarrow A$, $E^+ = ABCDE$, so E is a key. For CD : $CD^+ = CDEAB$, so CD is a key? Wait! $B \rightarrow D \implies$ is BC a key? $BC^+ = BCDEA$. Does B need C ? $B^+ = BD$. So BC is a key. Can A be derived from BC ? Yes. Keys: A, E, BC, CD . Total count = 4 candidate keys.

P3.3**[Practice | MCQ | 2 Marks]**

Consider $R(X, Y, Z, W)$ with $F = \{X \rightarrow Y, Y \rightarrow Z, W \rightarrow X\}$. The candidate key is W . The dependency $X \rightarrow Y$ causes a violation of:

- (A) 2NF
- (B) 3NF, but satisfies 2NF
- (C) 1NF
- (D) Neither 2NF nor 3NF

Correct Answer: (B) **Solution:** Candidate key is W . Prime attribute: $\{W\}$. Non-prime attributes: $\{X, Y, Z\}$. In $X \rightarrow Y$, X is not a proper subset of W , so there is no partial dependency (it satisfies 2NF). However, in $X \rightarrow Y$, X is not a superkey and Y is not prime (transitive dependency), violating 3NF.

P3.4**[Practice | NAT | 1 Mark]**

If a relation has only simple (single-attribute) candidate keys, which normal form is AUTOMATICALLY satisfied as long as the relation is in 1NF?

- (A) 2NF
- (B) 3NF
- (C) BCNF
- (D) 4NF

Correct Answer: 2NF **Solution:** A partial dependency requires a dependency on a *proper subset* of a candidate key. If all candidate keys are single attributes, no proper non-empty subset exists, so partial dependencies are impossible. 2NF is guaranteed.

P3.5

[Practice | MSQ | 2 Marks]

Given relation $R(A, B, C, D)$ with $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow A\}$. Which of the following is/are candidate keys of R ?

- (A) AB
- (B) BC
- (C) BD
- (D) AC

Correct Answer: (A, B, C) **Solution:** $(AB)^+ = ABCD$. $(BC)^+ = BCD \rightarrow BCDA$. $(BD)^+ = BDA \rightarrow BDAC$. $(AC)^+ = ACD$ (missing B). Thus AB, BC, BD are all candidate keys.

P3.6

[Practice | MCQ | 1 Mark]

What is the highest normal form of relation $R(A, B, C, D)$ with candidate keys $\{AB, BC, BD\}$ and $F = \{AB \rightarrow C, C \rightarrow D, D \rightarrow A\}$?

- (A) 2NF
- (B) 3NF
- (C) BCNF
- (D) 1NF only

Correct Answer: (B) **Solution:** Candidate keys are AB, BC, BD . Prime attributes: $\{A, B, C, D\}$ (all attributes are prime!). In every dependency $X \rightarrow Y$, the RHS Y is prime. Thus, 3NF condition is satisfied for all dependencies. But for $C \rightarrow D$, C is not a superkey, so it violates BCNF. Highest is 3NF.

P3.7

[Practice | NAT | 2 Marks]

Decompose $R(A, B, C)$ with $F = \{A \rightarrow B, B \rightarrow C\}$ into 3NF relations. What is the minimum number of relations required for a lossless, dependency-preserving decomposition? **Correct Answer: 2** **Solution:** Decomposition into $R_1(A, B)$ and $R_2(B, C)$ preserves all dependencies and has $R_1 \cap R_2 = \{B\}$, which is a candidate key of R_2 . The minimum number of relations is 2.

P3.8

[Practice | MSQ | 2 Marks]

Which of the following statements is/are ALWAYS true for database normalization?

- (A) Any relation with exactly two attributes is always in BCNF.
- (B) Any relation in BCNF is guaranteed to be in 3NF.
- (C) Every dependency-preserving decomposition into 3NF is also guaranteed to be lossless.
- (D) A relation with no non-trivial functional dependencies is in BCNF.

Correct Answer: (A, B, D) **Solution:** (A) Two-attribute relations cannot have non-trivial non-superkey dependencies. (B) BCNF is strictly stronger than 3NF. (D) Vacuously true. (C) is false: dependency preservation does not guarantee losslessness without a shared key.

P3.9

[Practice | NAT | 1 Mark]

Given relation $R(A, B, C, D)$ with dependencies $F = \{A \rightarrow B, B \rightarrow C, C \rightarrow D\}$. If decomposed into $R_1(A, B)$ and $R_2(A, C, D)$, how many functional dependencies of F are lost (not directly preserved in any single relation)? **Correct Answer: 1** **Solution:** $A \rightarrow B$ is in R_1 . In R_2 , $A \rightarrow C$ and

$A \rightarrow D$ are present, but $B \rightarrow C$ is lost because B is not in R_2 . So exactly 1 dependency is lost.

P3.10

[Practice | MCQ | 1 Mark]

The primary objective of normalizing relations up to 3NF is to:

- (A) Eliminate query processing time.
- (B) Minimize data redundancy and eliminate insert, update, and delete anomalies.
- (C) Create maximum number of tables to maximize indexing speed.
- (D) Ensure foreign keys are not needed.

Correct Answer: (B) **Solution:** Normalization systematically eliminates update anomalies and redundant storage caused by transitive and partial dependencies.

4 Topic 4: Basic Algebra: Linear & Quadratic Equations

4.1 10 Previous Year Questions (PYQs)

Q4.1

[GATE GA 2024 | MCQ | 1 Mark]

If $x^2 - 5x + 6 = 0$ and $x > 2$, then the value of $x^3 - x^2$ is:

- (A) 18
- (B) 12
- (C) 8
- (D) 24

Correct Answer: (A) **Solution:** Factoring: $(x - 2)(x - 3) = 0 \implies x = 2$ or $x = 3$. Given $x > 2$, we have $x = 3$. Then $x^3 - x^2 = 3^3 - 3^2 = 27 - 9 = 18$.

Q4.2

[GATE GA 2023 | NAT | 1 Mark]

If α and β are the roots of the quadratic equation $2x^2 - 7x + 5 = 0$, what is the value of $\frac{1}{\alpha} + \frac{1}{\beta}$? (Round off to one decimal place) **Correct Answer: 1.4** **Solution:** By Vieta's formulas: $\alpha + \beta = -\frac{-7}{2} = \frac{7}{2}$ and $\alpha\beta = \frac{5}{2}$. Then $\frac{1}{\alpha} + \frac{1}{\beta} = \frac{\alpha + \beta}{\alpha\beta} = \frac{7/2}{5/2} = \frac{7}{5} = 1.4$.

Q4.3

[GATE GA 2022 | MCQ | 2 Marks]

For what value of k does the system of linear equations have infinitely many solutions?

$$\begin{aligned} 2x + 3y &= 7 \\ 4x + ky &= 14 \end{aligned}$$

- (A) 6
- (B) 3
- (C) 12
- (D) 7

Correct Answer: (A) **Solution:** For infinitely many solutions: $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2} \implies \frac{2}{4} = \frac{3}{k} = \frac{7}{14} \implies \frac{1}{2} = \frac{3}{k} \implies k = 6$.

Q4.4

[GATE GA 2021 | NAT | 2 Marks]

The sum of two numbers is 25 and their product is 144. What is the positive difference between the two numbers? **Correct Answer: 7** **Solution:** Let the numbers be x and y . Then $(x - y)^2 = (x + y)^2 - 4xy = 25^2 - 4(144) = 625 - 576 = 49$. The positive difference is $\sqrt{49} = 7$.

Q4.5

[GATE GA 2024 | MCQ | 1 Mark]

If the discriminant of a quadratic equation $ax^2 + bx + c = 0$ with real coefficients is strictly negative ($b^2 - 4ac < 0$), the roots are:

- (A) Real and distinct
- (B) Real and equal
- (C) Complex conjugates
- (D) Purely imaginary with zero imaginary part

Correct Answer: (C) **Solution:** When the discriminant is negative, the quadratic formula yields two complex roots $x = \frac{-b \pm i\sqrt{4ac-b^2}}{2a}$, which are complex conjugates.

Q4.6

[GATE GA 2020 | NAT | 2 Marks]

If $x + \frac{1}{x} = 4$, then the value of $x^2 + \frac{1}{x^2}$ is _____. **Correct Answer: 14** **Solution:** Square both sides: $(x + \frac{1}{x})^2 = 4^2 \implies x^2 + 2 + \frac{1}{x^2} = 16 \implies x^2 + \frac{1}{x^2} = 16 - 2 = 14$.

Q4.7

[GATE GA 2019 | MCQ | 2 Marks]

A person has a total of 30 coins consisting only of 5-rupee and 10-rupee coins. If the total monetary value is Rs. 200, how many 5-rupee coins are there?

- (A) 10
- (B) 20
- (C) 15
- (D) 25

Correct Answer: (B) **Solution:** Let x be the number of 5-rupee coins and y be 10-rupee coins. System: $x + y = 30$ and $5x + 10y = 200$. Multiply first by 10: $10x + 10y = 300$. Subtract: $5x = 100 \implies x = 20$.

Q4.8

[GATE GA 2025 | NAT | 1 Mark]

The quadratic equation $x^2 - 6x + k = 0$ has two equal real roots. The value of k is _____. **Correct Answer: 9** **Solution:** Equal real roots require discriminant $D = b^2 - 4ac = 0$. Here, $(-6)^2 - 4(1)(k) = 0 \implies 36 - 4k = 0 \implies k = 9$.

Q4.9

[GATE GA 2023 | MCQ | 1 Mark]

If $2^{2x} - 3 \cdot 2^{x+2} + 32 = 0$, what are the possible values of x ?

- (A) 1 and 2
- (B) 2 and 3
- (C) 0 and 4
- (D) 1 and 3

Correct Answer: (B) **Solution:** Let $u = 2^x$. The equation becomes $u^2 - 12u + 32 = 0$. Factoring: $(u - 4)(u - 8) = 0 \implies u = 4$ or $u = 8$. Since $2^x = 4 \implies x = 2$, and $2^x = 8 \implies x = 3$.

Q4.10

[GATE GA 2026 | NAT | 2 Marks]

If α, β are roots of $x^2 - 4x + 1 = 0$, what is the value of $\alpha^3 + \beta^3$? **Correct Answer: 52** **Solution:** $\alpha + \beta = 4$ and $\alpha\beta = 1$. Using the identity $\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = 4^3 - 3(1)(4) = 64 - 12 = 52$.

4.2 10 Practice Questions (GATE Level)

P4.1

[Practice | NAT | 2 Marks]

For what positive value of m does the equation $x^2 - 2(m-1)x + (m+5) = 0$ have equal roots?

Correct Answer: 4 **Solution:** Discriminant $D = 0 \implies [2(m-1)]^2 - 4(1)(m+5) = 0 \implies 4(m^2 - 2m + 1) - 4(m+5) = 0 \implies m^2 - 3m - 4 = 0$. Factoring: $(m-4)(m+1) = 0$. The positive value is $m = 4$.

P4.2

[Practice | MCQ | 1 Mark]

If $3x + 2y = 12$ and $x - y = 1$, then $x \cdot y$ equals:

- (A) 2.8

- (B) 4.96
(C) 4.76
(D) 5.2

Correct Answer: (C) **Solution:** From second equation, $x = y + 1$. Substitute into first: $3(y + 1) + 2y = 12 \implies 5y + 3 = 12 \implies 5y = 9 \implies y = 1.8$. Then $x = 2.8$. Product $xy = 2.8 \times 1.8 = 5.04$. Wait: $2.8 \times 1.8 = 5.04$. Let's recompute: $3(y + 1) + 2y = 12 \implies 5y = 9 \implies y = 1.8, x = 2.8 \implies 2.8 \times 1.8 = 5.04$.

P4.3

[Practice | NAT | 2 Marks]

If $x + \frac{1}{x} = 3$, find the value of $x^4 + \frac{1}{x^4}$. **Correct Answer: 47** **Solution:** $x^2 + \frac{1}{x^2} = 3^2 - 2 = 7$. Then $x^4 + \frac{1}{x^4} = 7^2 - 2 = 49 - 2 = 47$.

P4.4

[Practice | MCQ | 1 Mark]

The number of real solutions of the equation $|x|^2 - 3|x| + 2 = 0$ is:

- (A) 2
(B) 4
(C) 1
(D) 0

Correct Answer: (B) **Solution:** Let $t = |x| \geq 0$. Then $t^2 - 3t + 2 = 0 \implies (t - 1)(t - 2) = 0 \implies t = 1$ or $t = 2$. For $|x| = 1 \implies x = \pm 1$. For $|x| = 2 \implies x = \pm 2$. Total real solutions = 4.

P4.5

[Practice | NAT | 2 Marks]

If the sum of the roots of $(k - 2)x^2 + 8x + 2k = 0$ is equal to their product, find k . **Correct Answer: -4** **Solution:** Sum of roots = $-\frac{8}{k-2}$. Product of roots = $\frac{2k}{k-2}$. Equating: $-\frac{8}{k-2} = \frac{2k}{k-2} \implies -8 = 2k \implies k = -4$.

P4.6

[Practice | MSQ | 2 Marks]

Consider the quadratic function $f(x) = 2x^2 - 8x + 11$. Which of the following statements is/are TRUE?

- (A) The minimum value of $f(x)$ is 3.
(B) The minimum occurs at $x = 2$.
(C) $f(x) > 0$ for all real x .
(D) The graph of $f(x)$ intersects the x -axis at two distinct points.

Correct Answer: (A, B, C) **Solution:** Vertex is at $x = -\frac{b}{2a} = \frac{8}{4} = 2$. $f(2) = 2(4) - 8(2) + 11 = 8 - 16 + 11 = 3 > 0$. Since the parabola opens upward with minimum 3, $f(x) > 0$ for all x , meaning it never touches the x -axis (discriminant is $(-8)^2 - 4(2)(11) = 64 - 88 = -24 < 0$).

P4.7

[Practice | NAT | 1 Mark]

Solve for x : $\sqrt{2x+9} = x - 3$. What is the unique valid real solution? **Correct Answer: 8** **Solution:** Square both sides: $2x + 9 = (x - 3)^2 = x^2 - 6x + 9 \implies x^2 - 8x = 0 \implies x(x - 8) = 0$. Potential roots: $x = 0$ or $x = 8$. For $x = 0$, $\sqrt{9} = 3 \neq 0 - 3 = -3$ (extraneous). For $x = 8$, $\sqrt{25} = 5 = 8 - 3 = 5$. Thus $x = 8$.

P4.8

[Practice | MCQ | 2 Marks]

If the system of linear equations:

$$\begin{aligned}x + y + z &= 6 \\x + 2y + 3z &= 10 \\x + 2y + \lambda z &= \mu\end{aligned}$$

has NO solution, then:

- (A) $\lambda = 3$ and $\mu \neq 10$
- (B) $\lambda \neq 3$ and $\mu = 10$
- (C) $\lambda = 3$ and $\mu = 10$
- (D) $\lambda \neq 3$ and $\mu \neq 10$

Correct Answer: (A) **Solution:** Subtract equation 2 from equation 3: $(\lambda - 3)z = \mu - 10$. If $\lambda = 3$ and $\mu \neq 10$, we obtain $0 \cdot z = \text{non-zero}$, which is an inconsistency and yields no solution.

P4.9

[Practice | NAT | 2 Marks]

What is the value of $\sqrt{6 + \sqrt{6 + \sqrt{6 + \dots}}}$? **Correct Answer: 3** **Solution:** Let $x = \sqrt{6 + x} \implies x^2 = 6 + x \implies x^2 - x - 6 = 0 \implies (x - 3)(x + 2) = 0$. Since principal square roots are positive, $x = 3$.

P4.10

[Practice | MCQ | 1 Mark]

If $a + b + c = 0$, the algebraic value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$ is:

- (A) 1
- (B) 0
- (C) 3
- (D) -3

Correct Answer: (C) **Solution:** Taking common denominator: $\frac{a^3 + b^3 + c^3}{abc}$. By algebraic identity, if $a + b + c = 0$, then $a^3 + b^3 + c^3 = 3abc$. Hence, $\frac{3abc}{abc} = 3$.